Reproducing NASA CARA's Monte Carlo Probability of Collision: An Open Two-Body Implementation, Checked on All 53 of CARA's Test Conjunctions

Sankur Kundu
Independent researcher, Bhubaneswar, India
sankur.kundu.tw@gmail.com

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Abstract

Collision-avoidance decisions for satellites rest on the probability of collision (Pc), most often computed with the two-dimensional method, which assumes a short, straight-line encounter. NASA's Conjunction Assessment Risk Analysis (CARA) team publishes 53 real conjunctions together with the result of its own two-body Monte Carlo for each, a reference for where the 2D method can be trusted. By CARA's own numbers it cannot be in 28 of the 53: the 2D value is off by more than a factor of two, too low in 25 and too high in 3, and by up to 162 orders of magnitude on slow, drifting encounters. We describe an open-source two-body Monte Carlo and test it against all 53 published results. On a two-proportion test of hit counts it agrees with CARA's within $|z| < 2$ on 50 conjunctions and $|z| < 3$ on all 53 ($\chi^2 = 48.0$ on 53 degrees of freedom, $p = 0.67$). Two modelling choices decide whether such a Monte Carlo agrees, and we measure each on the cases where it matters: the coordinates in which the Gaussian uncertainty is sampled, and the length of the time window, which for drifting encounters changes the answer by orders of magnitude. Run head to head with CARA's published library on the same inputs and one CPU core each, our implementation is 15–19$\times$ faster on encounters above 100 m/s, because each trial is searched only where a first-order bound says it can come within the hard-body radius. Code, inputs and raw results are released.[1]

1 Introduction

A satellite operator decides whether to manoeuvre from the probability that two objects, each known only through an estimated state and covariance, will pass within a combined hard-body radius (HBR) of each other. The method used most is the 2D probability of Foster and Estes [7], with refinements by Chan [4] and Alfano [1]: the relative motion near the time of closest approach (TCA) is taken as a straight line, the encounter as instantaneous, and the velocity as certain, so that the probability reduces to a two-dimensional Gaussian integral over a circle in the plane perpendicular to the relative velocity. These assumptions fail for slow encounters, long encounters, and encounters whose uncertainty is large enough that orbital curvature matters [6, 8].

CARA, which screens NASA's missions, addresses this with Monte Carlo methods [9, 11] and publishes, in its open CARA Analysis Tools [12], 53 conjunction data messages (CDMs) from real conjunctions together with its 2D, 3D and two-body Monte Carlo (SDMC) results for each. The test set was chosen to include the cases where the simpler methods break, and it is the natural benchmark for any independent implementation. This paper reports one, written from scratch, and makes three contributions:

1. A two-body Monte Carlo whose results are statistically indistinguishable from CARA's on all 53 test conjunctions (Section 3), reproducible from one script.

[1] https://github.com/Shakhtar-Sankur/kessler

2. Measurements of two modelling choices on the cases where each one decides the answer (Section 4). None of them is visible on a fast encounter, and an implementation can pass every fast case and still be wrong on the slow ones.
3. A search that restricts each trial to the time where it can hit, which makes the method 15–19× faster per core than CARA's library on most encounters (Section 5).

2 Method

Inputs. Each CDM gives both objects' positions and velocities at TCA in an Earth-centred inertial frame and their 6×6 covariances in each object's radial, transverse and normal (RTN) frame. We rotate each covariance to inertial axes with the block-diagonal RTN rotation, use CARA's published HBR for each case, and add nothing else: no drag or solar-pressure cross-correlation corrections, which CARA's published test values do not use either.

Sampling. The Gaussian must be taken to describe some set of coordinates, and the choice matters in the tails [10]. Our default is the equinoctial elements [3] $(n, a_f, a_g, \chi, \psi, \lambda)$, with the mean motion n as first element; the same elements with the semi-major axis a in place of n give the same results on the cases we tested (Section 4), and Cartesian sampling does not. The covariance is mapped from Cartesian coordinates through the Jacobian of the transformation, computed by central differences, and factored as $AA^\top$ by eigendecomposition, with the small negative eigenvalues that real CDM covariances contain set to zero. A trial draws $z_1, z_2 \sim N(0, I_6)$ and forms each object's elements as $\mu_i + A_i z_i$, then converts them back to a Cartesian state at TCA.

Motion and hits. Each sampled state is moved along its exact Kepler orbit using universal variables [13]; against numerical integration (relative tolerance 10^{-13}) the propagated positions agree to 10^{-4} m over $\pm$3,000 s on five test orbits, including retrograde and eccentric ones. A trial is a hit if the two objects come within the HBR at any time in the window. The minimum distance is found by stepping through the window at 60 s, locating each interval where the range rate changes sign from closing to opening, and refining the minimum there with a safeguarded Newton iteration on the range rate.

Window and which encounter. The window is half of the shorter orbital period either side of TCA, $\pm P/2$. Within it, only the close approach containing TCA is counted: two orbits that cross once usually cross again about half a revolution later, and where the times of possible approach separate into distinct intervals, the later one is a different conjunction.

Search windows. To first order, a trial's relative position at time t is the nominal one plus $M(t)z$, where $M(t)$ is the sensitivity of the relative position to the twelve standard normals, so it lies within $|z|\,\|M(t)\|_F$ of the nominal relative position. We tabulate $\|M(t)\|_F$ in 1 s cells by central differences along each column of A_1 and A_2, and keep the cells where the nominal distance can fall within the HBR for any $|z| \le 8$, with a margin of 1.5 times that reach plus 100 m for second-order terms. Trials with $|z| > 8$ (probability about 3×10^{-9} for twelve normals) are searched over the whole encounter instead, so that no trial is excluded on the strength of the bound alone. On synthetic encounters at 10 km/s, 300 m/s and 5 m/s, in all three sampling coordinates and for two radii, the restricted search gives exactly the hits of a full search, trial for trial (`tests/test_pc_mc.py`). On CARA's own conjunctions the same holds: for each of the 28 with a single region of possible approach within $\pm P/2$, the same 200,000 draws searched both ways give identical hit counts in 28 of 28 (929,154 hits in all, with the radius scaled up twentyfold so that every case has hits to compare; `bench/m3_windows_check.py`).

Randomness and statistics. Trials run in blocks of 65,536, each with its own xoshiro256** generator [2] seeded through splitmix64 from the run's seed and the block number, so a result depends on the seed and not on the number of threads (tested with 1 and 3). We report Clopper–Pearson 95% intervals [5] and compare

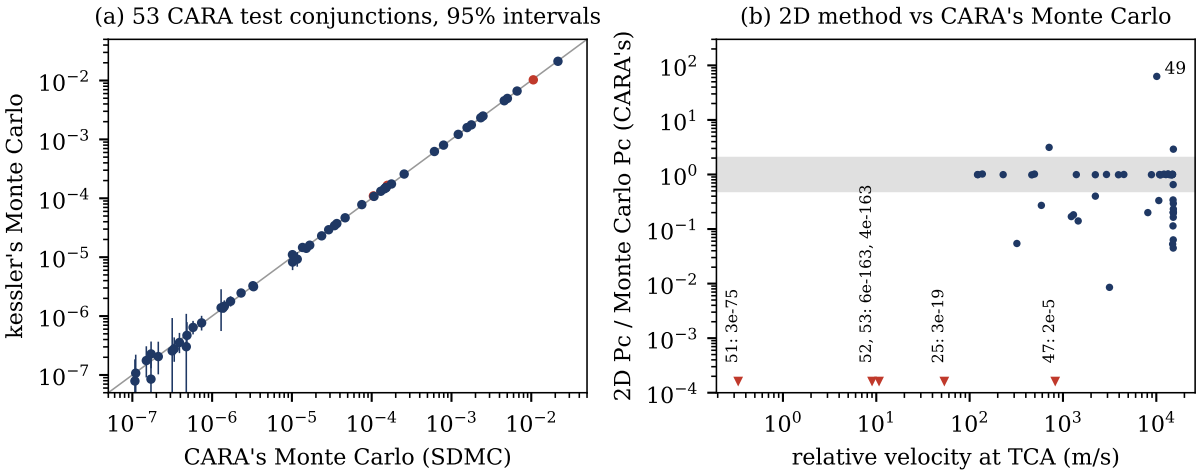


Figure 1: (a) Our Monte Carlo against CARA's on the 53 test conjunctions, with 95% intervals on both axes; red points have $|z| \geq 2$. (b) CARA's own 2D probability divided by its own Monte Carlo, against relative velocity at TCA; the grey band is a factor of two. Five cases fall below the axis and are marked with their ratios.

Table 1: Agreement with CARA's two-body Monte Carlo (SDMC) on its 53 test conjunctions.

Conjunctions with $\lvert z\rvert < 2$ / $\lvert z\rvert < 3$	50 / 53
Largest $\lvert z\rvert$	2.70
χ^2 over all 53 (53 degrees of freedom)	48.0 ($p = 0.67$)
Hits per case, ours: minimum / median	2 / 1196
Cases with fewer than 100 hits (ours)	17

with CARA's hits and trials by a two-proportion z test; over all cases, $\sum z^2$ is compared with a χ^2 distribution on 53 degrees of freedom.

3 Agreement with CARA on 53 conjunctions

For each conjunction we ran enough trials for about 3,000 hits, or as many as fitted in 120 s on 4 CPU cores: 1.79 billion trials in 29 minutes in all, against the 85.3 billion CARA ran. Figure 1a and Table 1 show the result. The largest $|z|$ is 2.70; 3 cases have $|z| \geq 2$, against 2.4 expected by chance in 53, and the sum of squares is what agreement predicts. Our implementation of the 2D method itself matches CARA's 2D values to a relative difference of at most 1.6×10^{-8}.

Table 2 shows the cases that make the test set hard. The 2D method underestimates case 25 by a factor of 3×10^{18} and the two drifting cases 52 and 53 by more than 10^{160}: at 9–11 m/s the objects stay close for a large fraction of an orbit, nothing about the encounter is short or straight, and the 2D integral sees only a vanishing tail. It also overestimates: case 49 by a factor of 63. Figure 1b shows the pattern over all 53. The failures are not confined to slow encounters: of the 27 conjunctions above 10 km/s, 14 are off by more than a factor of two, up to 22 times too low and 63 times too high. The Monte Carlo agrees with CARA's on these as on the others.

The test has limited power where hits are few. For the conjunctions with Pc $\sim 10^{-7}$, 120 s bought only tens of millions of trials, and 17 cases have fewer than 100 hits (as few as 2). On those the comparison detects only gross disagreement; the conjunctions that discriminate between modelling choices are run with far more trials in the next section.

Table 2: Selected conjunctions: the 2D method, CARA's Monte Carlo and ours.

Case	Objects	v_{rel}	2D (CARA)	CARA SDMC	This work	z
1	Hubble, Delta 2 rocket body	2.9 km/s	6.11×10^{-4}	6.11×10^{-4}	6.23×10^{-4}	$+0.9$
25	WorldView-2, Fengyun-1C debris	53.6 m/s	4.5×10^{-23}	1.51×10^{-4}	1.53×10^{-4}	$+0.5$
45	ICESat-2, SpaceBEE-114	15.2 km/s	3.3×10^{-8}	7.38×10^{-7}	7.7×10^{-7}	$+0.3$
49		10.1 km/s	2.4×10^{-5}	3.89×10^{-7}	3.6×10^{-7}	-0.5
51		0.3 m/s	4.5×10^{-81}	1.30×10^{-6}	1.4×10^{-6}	$+0.2$
52	TROPICS Pathfinder, LINCS2	9.0 m/s	6.5×10^{-168}	1.17×10^{-5}	9.3×10^{-6}	-1.6
53		10.7 m/s	3.9×10^{-168}	1.02×10^{-5}	8.3×10^{-6}	-1.4

Table 3: Sampling coordinates, at the default window ($\pm P/2$, encounter at TCA only). z is against CARA's hits and trials.

Case	Gaussian in	Pc, this work	Hits / trials	CARA SDMC	z
1	Cartesian state	6.08×10^{-4}	3,040 / 5M	6.11×10^{-4}	-0.2
1	equinoctial, a	6.17×10^{-4}	3,087 / 5M	6.11×10^{-4}	$+0.5$
1	equinoctial, n	6.13×10^{-4}	3,063 / 5M	6.11×10^{-4}	$+0.1$
25	Cartesian state	1.27×10^{-4}	509 / 4M	1.51×10^{-4}	-3.7
25	equinoctial, a	1.57×10^{-4}	628 / 4M	1.51×10^{-4}	$+1.0$
25	equinoctial, n	1.53×10^{-4}	614 / 4M	1.51×10^{-4}	$+0.5$
45	Cartesian state	0	0 / 100M	7.38×10^{-7}	-8.6
45	equinoctial, a	7.60×10^{-7}	76 / 100M	7.38×10^{-7}	$+0.3$
45	equinoctial, n	9.80×10^{-7}	98 / 100M	7.38×10^{-7}	$+2.8$
45	equinoctial, a	7.58×10^{-7}	758 / 1,000M	7.38×10^{-7}	$+0.7$
45	equinoctial, n	7.40×10^{-7}	740 / 1,000M	7.38×10^{-7}	$+0.1$

4 Two choices that decide agreement

Sampling coordinates. A linear covariance describes a Gaussian in some coordinates, and its image in other coordinates is not Gaussian; the difference grows with the size of the along-track uncertainty [10]. Table 3 draws the same three conjunctions in three ways: Cartesian states, equinoctial elements with the semi-major axis a, and equinoctial elements with the mean motion n. On case 1, a fast encounter with small uncertainty, all three agree with CARA. On case 25 Cartesian sampling is 15% low ($z = -3.7$): a straight-line draw in position and velocity tilts each sample's velocity off its orbit, and at 54 m/s that changes how long the objects stay close. On case 45, whose probability of 7.4×10^{-7} is decided by the far tail, Cartesian sampling finds no hit at all in 10^8 trials, where CARA's value predicts about 74. Equinoctial sampling, in either form, reproduces that tail: with 10^9 trials each, the semi-major-axis form gives 7.58×10^{-7} ($z = +0.7$) and the mean-motion form 7.40×10^{-7} ($z = +0.1$), against CARA's 7.38×10^{-7}, and the two are indistinguishable at this precision (about $\pm 4\%$). At 10^8 trials the mean-motion form had read 33% high ($z = +2.8$, 98 hits), and an earlier, smaller run had suggested the semi-major-axis form was high; neither survived ten times the trials. With around a hundred hits, an excursion of two to three standard errors somewhere in a table of comparisons is to be expected, and a claim about the far tail needs hits in the hundreds.

Window length and which encounter. CARA's standalone `Pc_SDMC` wrapper defaults to a window of a quarter of the shorter period either side of TCA [12]. That is ample for fast encounters and too short for drifting ones (Table 4): on cases 52 and 53, at 9 and 11 m/s, $\pm P/4$ finds no hit in 8×10^6 trials where CARA's values predict about 90, because the objects approach so slowly that the hits happen more than a quarter period from TCA. At $\pm P/2$ both agree with CARA's. At $\pm P$ both come out 40–50% higher ($z \approx +4.4$), and counting only the approach containing TCA does not change that: at these speeds the times at which the two objects can come within reach form one unbroken span, so there is no separate second encounter to exclude. For a drifting pair, the probability of collision "in this conjunction" is not defined independently of the window, and the window must be reported with the number. CARA's published values

Table 4: Window length and which close approach is counted, on the two drifting conjunctions.

Case	Window	Counted	Pc, this work	Hits	CARA SDMC	z
52	$\pm P/4$	TCA only	0	0	1.17×10^{-5}	-9.7
52	$\pm P/2$	TCA only	1.26×10^{-5}	101	1.17×10^{-5}	$+0.8$
52	$\pm P$	TCA only	1.70×10^{-5}	136	1.17×10^{-5}	$+4.4$
52	$\pm P$	every approach	1.70×10^{-5}	136	1.17×10^{-5}	$+4.4$
53	$\pm P/4$	TCA only	0	0	1.02×10^{-5}	-9.0
53	$\pm P/2$	TCA only	1.10×10^{-5}	88	1.02×10^{-5}	$+0.7$
53	$\pm P$	TCA only	1.52×10^{-5}	122	1.02×10^{-5}	$+4.4$
53	$\pm P$	every approach	1.52×10^{-5}	122	1.02×10^{-5}	$+4.4$

Table 5: Time for 10^6 trials on one core, CARA's library and this work, same inputs.

Case	v_{rel} (m/s)	CARA library (s)	This work (s)	Ratio
5	14,508	31.0	1.64	18.9×
2	11,073	31.4	1.69	18.6×
22	4,489	29.1	1.61	18.1×
1	2,925	29.1	1.94	15.0×
8	1,390	31.5	1.68	18.7×
14	123	31.2	1.84	17.0×
25	54	31.2	17.66	1.8×

for these two cases correspond to $\pm P/2$ in our implementation; we do not know the span CARA used to produce them. Where the times of possible approach do separate, in 25 of the 53 conjunctions within $\pm P/2$, our implementation counts only the interval containing TCA.

5 Speed

CARA distributes its two-body Monte Carlo as a compiled library (`libsdmc_linux.so`, built with Intel Fortran) with a C++ entry point [12]. We built a driver around the published library and ran it and our implementation on the same machine with the same states, covariances, HBR and window ($\pm P/4$, the library's default), one core each and 10^6 trials per case (`bench/m3_vs_nasa_lib.py`). Both give the same probabilities within statistical error (largest $|z|$ 2.42).

On encounters from 123 m/s to 14.5 km/s our implementation takes 1.6–1.9 s per million trials where the library takes 29–31 s (Table 5). Most of the difference comes from the search windows of Section 2: a fast encounter can hit only in a few seconds around TCA, and a trial needs to be propagated only there. On the slowest case run, 25 at 54 m/s, the window that can hit spans most of the revolution, and the advantage falls to 1.8×. We did not study the library's internals, and the comparison measures the two programs as distributed, on one machine.

6 Limitations

The method is two-body, as CARA's SDMC is; it does not model drag, solar pressure or manoeuvres between the CDM epoch and the encounter, nor the cross-correlation corrections CARA can apply. The Gaussian is a modelling assumption in whichever coordinates it is placed, and the agreement shown here is agreement with CARA's choices, not with ground truth, which no one has for these conjunctions. The 60 s step of the minimum-distance search assumes no two separate minima of the range within one step. The search-window bound is first-order with a margin; it changed no hit on synthetic encounters or on 28 of CARA's conjunctions, but it is not proven. The speed comparison is one machine and seven conjunctions. For drifting encounters the probability depends on the time window, and agreement on cases 52 and 53 is agreement at $\pm P/2$.

7 Reproducing the results

`make && python bench/m3_sdmc.py` reproduces Section 3; `python bench/m3_ablations.py` and `bench/m3_tail.py` Section 4; `python bench/m3_windows_check.py` the search-window check; `bench/m3_vs_nasa_lib.py` Section 5 (with a checkout of CARA Analysis Tools and Intel's Fortran runtime); `python paper/figures.py` regenerates every number, table and figure in this paper from the results files. The CDMs are fetched from CARA's repository at a pinned commit and are not redistributed.

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